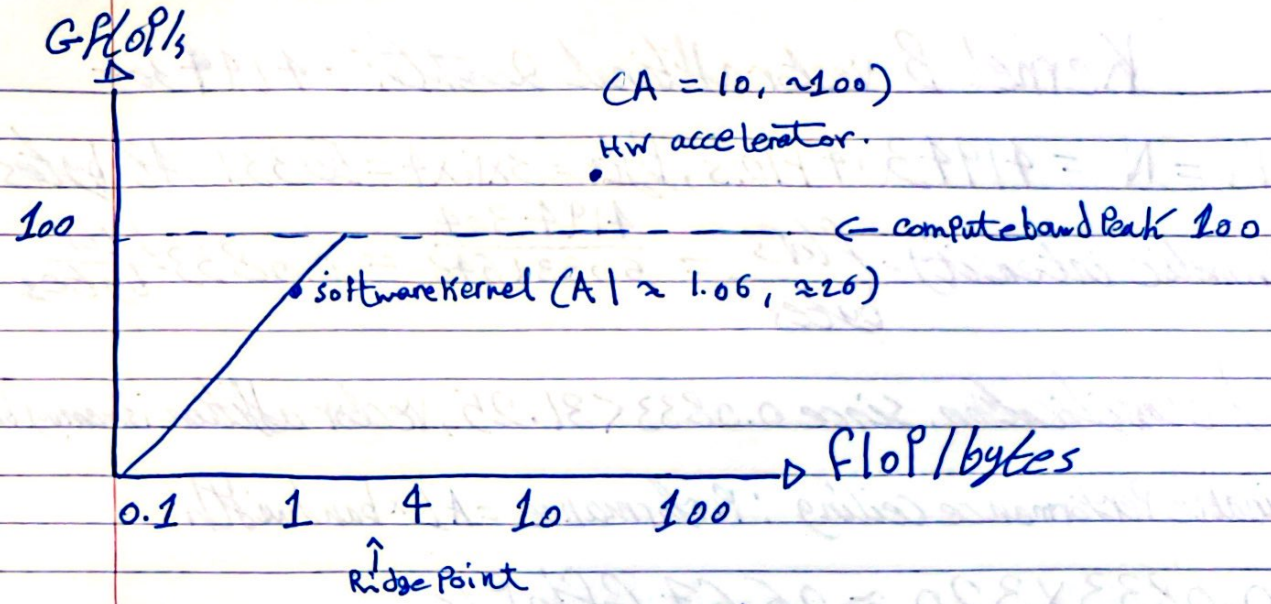
Codefest 2 CMAN Roofline Dawood



Hardware: Peak compute = 10 TFLOPS = 10,000 GFLOPS

Peak memory bandwith = 320 GB/s, Ridge Point = $\frac{10K}{320}$ = 31.25 FLOP/byte

Kernel A

FLOPs = $2 \times N^3 = 2 \times 1024^3 = 2147483648$ FLOPs

Bytes = $3 \times N^2 \times 4 = 3 \times 1024^2 \times 4 = 12582912$ bytes.

Arithmetic intensity = FLOPs/bytes $\Rightarrow \frac{2147483648}{12582912} \approx 170.67 \frac{FLOP}{bytes}$

Bound classification: since 170.67 > 31.25, GEMM is compute-bound.

Attainable Performance ceiling, compute ceiling = 10000 GFLOPS

Architectural recommendation. Increase compute Throughput, such as adding more MAC units or wider matrix hardware, because GEMM is already above the ridge point.

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Kernel B vector addition length = 4194304

FLOPs = N = 4194304 FLOPs, Bytes = 3×N×4 = 50331648 bytes.

Arithmetic intensity = $\frac{\text{flops}}{\text{bytes}} = \frac{4194304}{50331648} = 0.0833 \frac{\text{flop}}{\text{bytes}}$

Bound classification. since 0.0833 < 31.25, vector addition is memory bound.

Attainable Performance ceiling : Performance = AI × bandwidth

$= 0.0833 \times 320 \approx 26.67$ GFLOPs.

Architectural recommendation.

Increase memory bandwidth or reduce DRAM traffic, because vector addition is limited by data movement.

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